Ireland HSE Outpatient Waiting List Analytics & Forecasting Dashboard

Data Analytics Portfolio Project

Python | PostgreSQL | Power BI | Machine Learning

Data Source: National Treatment Purchase Fund (NTPF) Ireland

Prepared by

Monci Mamachan

Introduction

Ireland's healthcare waiting list crisis is one of the most pressing public health challenges facing the country. This project analyses real government data from the National Treatment Purchase Fund (NTPF) to uncover patterns, identify the most critically underperforming hospitals, and forecast future demand.

Using 25 months of real Irish government data covering 44 hospitals from January 2024 to January 2026, this project delivers actionable insights through a complete data analytics pipeline — from raw SQL data engineering through to machine learning forecasting and interactive Power BI dashboards.

Key Statistics

631,608	49,373	44	628,120
Patients Waiting Jan 2026	Waiting 18+ Months	Hospitals Analysed	Forecast by July 2026

Project Tools

Tool	Purpose
PostgreSQL 17	Data ingestion, cleaning, feature engineering, analytical queries
Python (Pandas)	Exploratory data analysis and feature engineering
Matplotlib / Seaborn	4 professional data visualisation charts
Scikit-learn	Linear Regression forecasting model
Power BI Desktop	3-page interactive dashboard for stakeholder reporting
NTPF Open Data	Real Irish government hospital waiting list data

Key Findings

Finding 1 — The Crisis Is Getting Worse

Ireland's total outpatient waiting list grew from 568,691 patients in January 2024 to 631,608 in January 2026 — an increase of 62,917 patients (+11%) in just two years. January 2026 recorded the largest single-month increase in the entire dataset at +3.21%.

Month	Total Waiting	Change
January 2024	568,691	Baseline
December 2024	556,248	-2.2% (best month)
August 2025	623,008	Summer peak
January 2026	631,608	+3.21% (worst ever)

Finding 2 — Progress Was Made Then Lost

2024 showed genuine improvement — critical patients waiting 18+ months fell from 54,804 to 42,996 (a 21% reduction). However all of this progress was reversed in 2025, with critical waiting rising back to 49,373 by January 2026. The January 2026 critical spike of +11.83% is the most alarming data point in the entire analysis.

Finding 3 — Hidden Crisis Hospitals

Total waiting list size is misleading. The most critical metric is the percentage of patients waiting over 18 months. Several smaller hospitals have catastrophically high crisis rates that are invisible in volume-based rankings:

Hospital	Total Waiting	18 Month+ %	Status
St. Columcille's Hospital	10,146	55.05%	🔴 CRITICAL
Nenagh Hospital	1,270	25.04%	🔴 CRITICAL
Ennis Hospital	1,364	23.39%	🔴 CRITICAL
South Infirmary Victoria UH	34,130	15.45%	🟠 HIGH
National Orthopaedic Hospital Cappagh	2,812	13.19%	🟠 HIGH
Galway University Hospitals	50,024	12.91%	🟠 HIGH
Cork University Hospital	25,167	1.34%	🟢 GOOD
Rotunda Hospital	6,010	0.03%	🟢 EXCELLENT

Finding 4 — Seasonal Patterns

A clear seasonal pattern exists across both years. Waiting lists consistently rise from January through August as hospitals slow down over summer while referrals continue. They then fall from September through December as capacity increases. This pattern has direct implications for resource planning and budget allocation.

Technical Approach

Layer 1 — PostgreSQL Data Engineering

- **Schema Design:** Designed a normalised 3-table schema matching the exact NTPF CSV structure including handling of comma-formatted numbers
- **Data Cleaning:** 5 quality checks including NULL detection, duplicate identification, date standardisation and format conversion
- **Analytical Queries:** 3 advanced SQL queries using CTEs (Common Table Expressions), window functions (LAG, DATE_TRUNC), and subqueries
- **Key SQL Skills:** GROUP BY aggregations, HAVING clauses, EXTRACT, ROUND, CASE WHEN, rolling window calculations

Layer 2 — Python Exploratory Data Analysis

- **Libraries:** Pandas for data manipulation, Matplotlib and Seaborn for visualisation, NumPy for array operations
- **Chart 1:** National trend line showing total and critical waiting patients across 25 months on dual axes
- **Chart 2:** Horizontal bar chart of top 10 hospitals by volume with critical patient overlay
- **Chart 3:** Hidden crisis chart — all 44 hospitals ranked by 18 month+ percentage with colour-coded severity tiers and national average reference line
- **Chart 4:** 6-month forecast visualisation showing historical data connected to regression predictions

Layer 3 — Forecasting Model

- **Algorithm:** Linear Regression (Scikit-learn) trained on 25 months of national monthly totals
- **Features:** Month number as sequential integer predictor — simple, explainable, and effective for trend projection
- **Output:** Monthly growth rate of 1,821 patients per month with 6-month forward projections
- **Forecast:** Predicts 628,120 patients on waiting lists by July 2026 if current trends continue

Layer 4 — Power BI Dashboard

- **Page 1 — National Overview:** KPI cards, monthly trend line, top 10 hospital bar chart, adult/child slicer
- **Page 2 — Hospital Risk Analysis:** Crisis bar chart with conditional colour formatting, scatter risk map, full hospital table with red/green scale
- **Page 3 — Forecast:** 6-month forecast line chart connecting actual data to predictions with KPI cards

Policy Recommendations

These recommendations are derived directly from the data analysis and are intended for HSE healthcare planners and policy makers.

Priority	Recommendation	Target	Data Evidence
1 — URGENT	Immediate intervention at St. Columcille's Hospital	St. Columcille's	55% of patients waiting 18+ months
2 — URGENT	Emergency capacity review at Nenagh and Ennis Hospitals	Nenagh, Ennis	25% and 23% crisis rates respectively
3 — HIGH	Seasonal resource planning — increase capacity Jan-Aug	All hospitals	Consistent summer surge every year
4 — HIGH	Replicate Cork University Hospital best practices	Underperforming hospitals	CUH maintains 1.34% crisis rate
5 — MEDIUM	Monthly forecasting model to predict demand spikes	HSE Planning Team	Jan 2026 spike was predictable from trend
6 — ONGOING	Track 18 month+ rate as primary KPI not total volume	All reporting	Volume metrics hide hidden crisis hospitals